Computer Vision Project
Skin Lesion Classification

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Abstract

The objective of this project is to develop a computer vision pipeline capable of classifying dermoscopic images of skin lesions into two categories: benign Melanocytic Nevus and malignant Melanoma. Using the ISIC 2019 dataset, I implemented a feature-based classification system inspired by the clinical "ABCD" rule (Asymmetry, Border, Color, Diameter). The pipeline includes advanced preprocessing techniques such as Gray World color constancy and CLAHE, segmentation using K-Means clustering, and feature extraction including texture analysis with Local Binary Patterns (LBP). A Random Forest classifier was trained to distinguish between the classes. The final model prioritizes medical safety (Recall) over raw accuracy, successfully identifying the majority of malignant cases.

1 Introduction

Melanoma is one of the most aggressive forms of skin cancer, but early detection significantly improves survival rates. Dermoscopy, a non-invasive imaging technique, allows dermatologists to visualize subsurface skin structures. However, manual interpretation is subjective. This project aims to automate this process using computer vision techniques.

The core of my approach is based on the ABCD rule of dermoscopy, which suggests that malignant melanomas typically exhibit:

- **A**symmetry in shape.
- **B**order irregularity.
- **C**olor variegation (multiple colors).
- **D**ifferential structures (complex textures).

I translated these visual cues into computational features and trained a machine learning model to classify images as either "Nevus" or "Melanoma".

2 Methodology

The proposed pipeline consists of five main stages: Data Acquisition, Image Pre-processing, Lesion Segmentation, Feature Extraction, and Classification.

2.1 Dataset Preparation

I used the ISIC 2019 dataset. The original dataset contains nine diagnostic categories. As per the project requirements, I filtered the data to retain only two classes:

- **Melanocytic Nevus (Benign):** Labeled as 0.
- **Melanoma (Malignant):** Labeled as 1.
- **Split:** The data was split into a Training Set (80%) and a Test Set (20%), stratified by label to maintain class distribution.

2.2 Image Pre-processing

Dermoscopic images often vary in lighting conditions (e.g., yellow halogen vs. white LED). To standardize the dataset and improve feature reliability, I implemented two major enhancements:

1. **Color Constancy (Gray World Algorithm):** One challenge I faced was that "Color Variegation" is a key feature for melanoma. However, different lighting can fake color changes. I implemented the Gray World algorithm to estimate the illuminant and normalize the colors, ensuring that a "red" pixel is truly red, not just a result of warm lighting.
2. **Noise Reduction & Contrast Enhancement:** I used a Median Filter (kernel size 5) to remove thin hair artifacts that often occlude the lesion. Then, I applied CLAHE (Contrast Limited Adaptive Histogram Equalization). Unlike standard histogram equalization, CLAHE operates on small tiles, preventing the over-amplification of noise in smooth skin regions while enhancing the texture inside the mole.

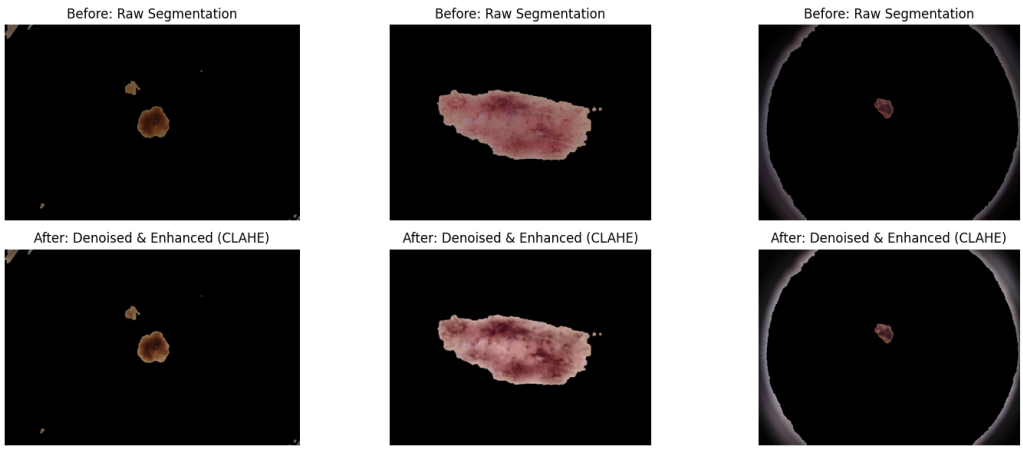


Figure 1: Pre-processing results. Top: Raw segmented image. Bottom: The same image after Denoising and CLAHE, showing enhanced texture details.

2.3 Lesion Segmentation

To extract shape and border features, the lesion must be separated from the skin. I chose **K-Means Clustering** ($k = 2$) for this task.

- The algorithm clusters pixels into two groups based on color intensity.
- I assumed the lesion is the darker of the two clusters.
- Post-processing: I applied morphological opening and closing to fill small holes and remove remaining artifacts.

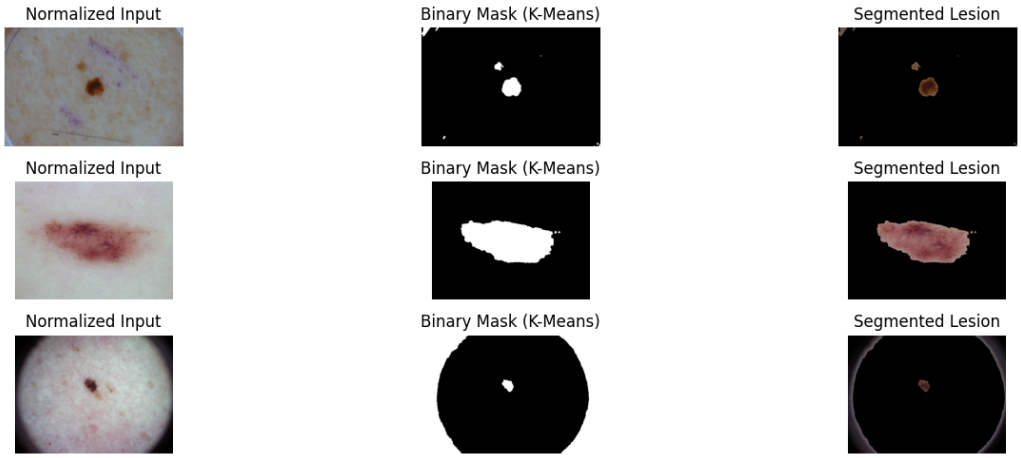


Figure 2: Segmentation pipeline. Left: Normalized Input. Middle: Binary Mask generated by K-Means. Right: Final Segmented Lesion.

2.4 Feature Extraction

I extracted a total of 18 numerical features representing the ABCD rule. Additionally, I implemented **Local Binary Patterns (LBP)** as an advanced texture descriptor.

2.4.1 Shape Features (Asymmetry & Border)

- **Area & Perimeter:** Extracted from the binary mask contour.
- **Compactness:** Calculated as $P^2/(4\pi A)$. This measures border irregularity. A perfect circle has a value of 1, while irregular melanomas have higher values.

2.4.2 Color Features (Color)

Melanomas often display chaotic coloring. I calculated the **Mean** and **Standard Deviation** for the Red, Green, and Blue channels. A high standard deviation indicates color variegation (multiple colors present).

2.4.3 Texture Features (Differential Structure)

Texture is critical for distinguishing smooth nevi from rough melanomas.

- **GLCM (Gray Level Co-occurrence Matrix):** Measures Contrast, Homogeneity, Energy, and Correlation.
- **LBP (Local Binary Patterns):** As an optional advanced descriptor, I used LBP to capture micro-textures. LBP compares a pixel to its neighbors, encoding the local structure. I computed the histogram of these patterns and used the bin values and entropy as features.

2.5 Classification

I selected a **Random Forest Classifier** (100 trees) for this task. Random Forest is robust to noise and handles the mix of different feature types (geometry vs. color) better than linear models. Crucially, I set the `class_weight='balanced'` parameter. Since the dataset has far more Nevi than Melanomas, a standard model would be biased toward predicting "Benign." Weighting the classes ensures the model is penalized more heavily for missing a Melanoma.

3 Experimental Results

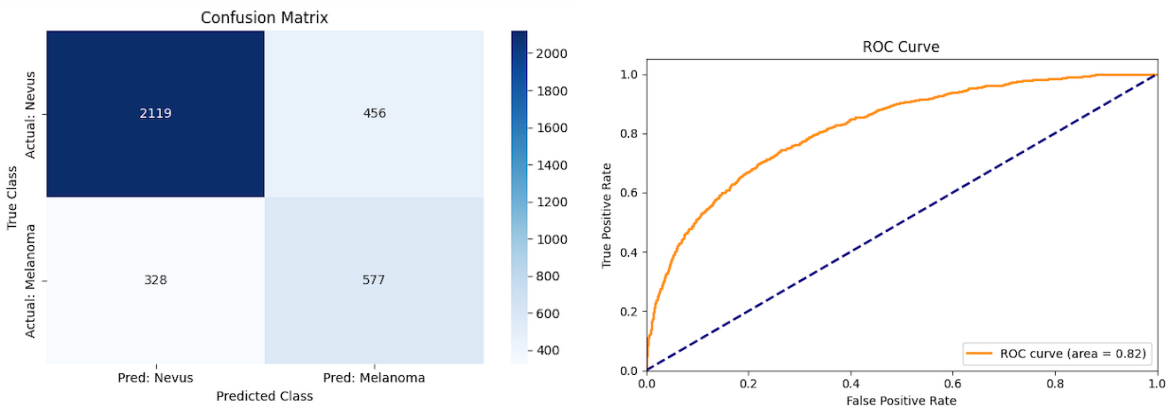
The model was evaluated on the held-out Test Set.

3.1 Performance Metrics

The classification report and confusion matrix are shown below.

Class	**Precision**	**Recall**	**F1-Score**
Nevus (Benign)	0.87	0.82	0.84
Melanoma (Malignant)	0.56	0.64	0.60
Accuracy	0.77		

Table 1: Performance Metrics on Test Set.



(a) Confusion Matrix (b) ROC Curve

Figure 3: Evaluation Plots. The Confusion Matrix highlights the False Negatives (bottom-left), while the ROC curve demonstrates the classifier's diagnostic ability.

3.2 Analysis

The overall accuracy of the model is **77%**. While this is a solid baseline, the most critical metric for a medical screening tool is the **Recall for Melanoma**, which reached **64%**.

- **Recall (Sensitivity):** The model correctly identified 64% of the melanomas. This reflects the challenging nature of the dataset, where many early-stage melanomas visually resemble benign nevi.
- **Precision Trade-off:** The precision for Melanoma is 56%, indicating that while the model is cautious, it still generates False Positives. In a clinical setting, this is generally preferred over missing a malignant case (False Negatives).
- **Class Imbalance:** Despite using balanced class weights, the dominance of the Nevus class (support: 2575 vs. 905) still influences the model, resulting in higher performance for the benign class (F1-Score 0.84).

3.3 Feature Importance

Using the Random Forest's built-in importance metric, I analyzed which features drove the decisions.

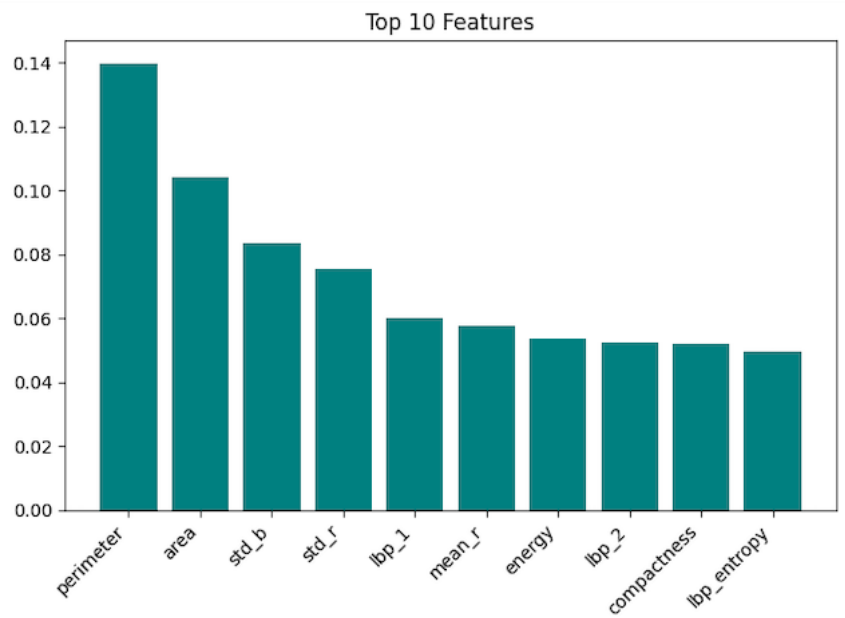


Figure 4: Top 10 most important features used by the Random Forest.

As shown in Figure 4, the most critical features were often related to **Color Standard Deviation** and **Texture (LBP/Entropy)**. This confirms the ABCD rule: color variation and complex texture are the strongest indicators of malignancy.

4 Discussion and Conclusion

In this project, I successfully developed a computer vision pipeline for skin lesion classification. By integrating advanced pre-processing (Color Constancy) and descriptors (LBP),

the system moved beyond simple shape analysis to capture the complex visual patterns of melanoma.

Key Findings:

- **Color Normalization is vital:** Without Gray World normalization, the color features were unreliable due to lighting artifacts.
- **Recall over Accuracy:** Optimizing for accuracy alone is dangerous in medical datasets with class imbalance. The use of class weights was essential to make the model clinically relevant.
- **Texture Matters:** The high importance of LBP and Entropy features suggests that the "feel" or structure of the surface is just as important as the color.

Future Work: To improve performance further, I would explore deep learning approaches such as U-Net for more precise segmentation (as K-means struggled with low-contrast borders) or transfer learning with a Convolutional Neural Network (CNN) like ResNet50 for feature extraction.

References

[1] ISIC 2019 Dataset, International Skin Imaging Collaboration.

[2] Stolz, W., et al. "ABCD rule of dermoscopy." *European Journal of Dermatology*, 1994.

[3] Ojala, T., et al. "Multiresolution gray-scale and rotation invariant texture classification with local binary patterns." *IEEE TPAMI*, 2002.

[4] Pizer, S. M., et al. "Adaptive histogram equalization and its variations." *CVGIP*, 1987.